Monthly Intelligence Report

Edition 001 — The Osaka–Aichi Corridor

Where the energy transition is outrunning the grid. Inaugural edition · April 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED

5,981

299 EHV · 5,682 HV

GRID LINES MAPPED

~11,600

~105,000 km coverage

MC SIMULATIONS

78.9 M

10,000 per substation

PUBLIC DATA SOURCES

30+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 78.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 469,404 output data points with full uncertainty quantification across 5,981 substations and ~11,600 grid lines spanning ~105,000 km.

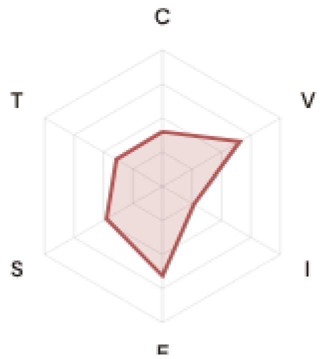
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering Japan's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

一関変電所

Miyagi, Tohoku · 154 kV

0.607	High	0.307	67th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

C Continuity: 0.399 / 0.30 (133%)

I Infrastructure: 0.265 / 0.25 (106%)

S Saturation: 0.474 / 0.20 (237%)

V Voltage: 0.657 / 0.10 (657%) ⚠

E Economic: 0.656 / 0.10 (656%) ⚠

T Transition: 0.390 / 0.05 (780%)

MODIFIERS

R3 Consequence: 1.055 ↑ amplifies

R4 Graph Criticality: 1.223 ↑ amplifies

R6a Restoration: 1.009 → neutral

R6b Seismic: 1.078 ↑ amplifies

R7 Digital Readiness: 1.001 → neutral

This month's spotlight — automatically selected for April 2026 — is **一関変電所** in Miyagi, Tohoku. With an R_final of 0.607 (High band, 67th fleet percentile), its risk profile is dominated by the T (Transition) component at 780% of its weight ceiling, followed by V (Voltage) at 657%. Modifiers collectively shift R_base by 40.6, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Society 5.0 / Japan 2030 digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

2236

High/Critical + R7 < median

AVG R7 MODIFIER

1.0012

Range [0.99, 1.05]

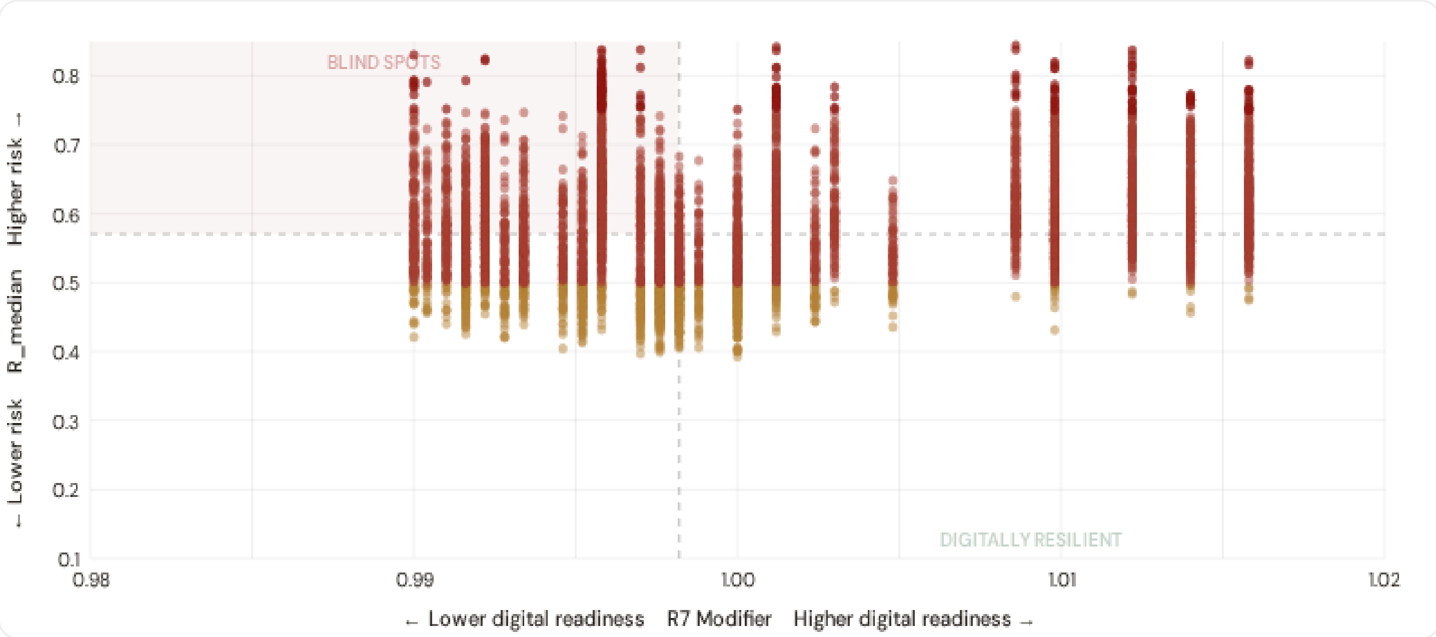
HIGH-CONSEQUENCE SUBS

2,431

40.6% · R3 ≥ 1.12

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



• Low • Medium • High • Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **2236 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 0.9982). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Tohoku (686), Hokkaido (403), Kyushu (335)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier — indicating strong digital infrastructure coverage but concentrated in the southern urban grid.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: Est. high VoLL

Heavy industry, critical infrastructure -- highest economic loss per interruption

2,193 substations (36.7%) High/Critical: **2158** (98.4%) Avg R: **0.626** Avg E: **0.553** / 0.10

Low 0 Med 35 High 2001 Crit 157

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: Est. medium-high VoLL

Diversified industry, logistics, processing -- significant continuity requirements

800 substations (13.4%) High/Critical: **744** (93.0%) Avg R: **0.594** Avg E: **0.672** / 0.10

Low 0 Med 56 High 711 Crit 33

🏢 Commercial / SME-Dense

Est. VoLL: Est. medium VoLL

Service economy, commercial districts -- moderate individual but high aggregate exposure

1,533 substations (25.6%) High/Critical: **1165** (76.0%) Avg R: **0.551** Avg E: **0.689** / 0.10

Low 0 Med 368 High 1160 Crit 5

🌾 Light / Agricultural / Remote

Est. VoLL: Est. low VoLL

Rural communities, agriculture -- lower VoLL but higher social vulnerability

1,455 substations (24.3%) High/Critical: **888** (61.0%) Avg R: **0.527** Avg E: **0.661** / 0.10

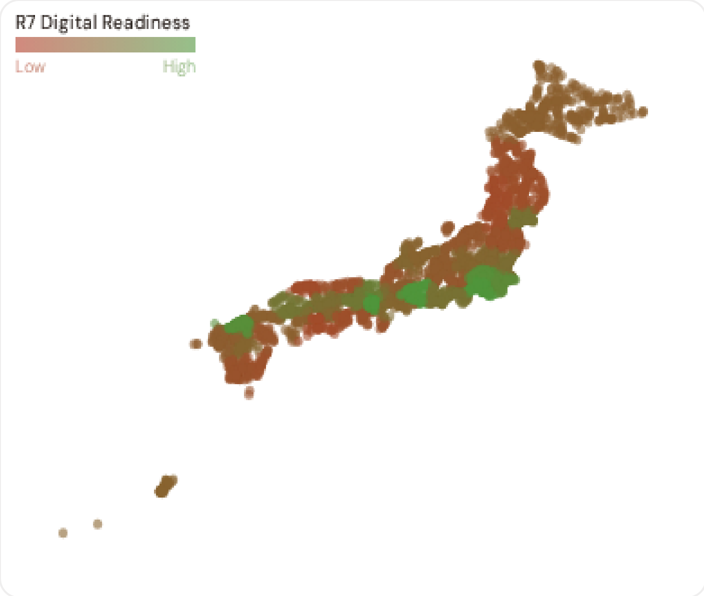
Low 0 Med 567 High 881 Crit 7

Value of Lost Load (VoLL) context: METI's 2024 estimate for Japan places average VoLL at ¥17/kWh for industrial customers and ¥4.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **2158 substations** combine capital-intensive economic fabric (R3 ≥ 1.14) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **TEPCO (1068), Hokkaido (403), Kansai (386)** — regions where industrial corridors depend on uninterrupted power for continuous processes (steel, glass, automotive). A single hour of unplanned outage at these substations carries an estimated VoLL of ¥20–40/kWh, compared to ¥1–4/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 8.0%, but this masks sharp regional variation: Remote and northern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Prefecture Digital Readiness Ranking

Prefectures ranked by average R7 modifier — lower values indicate weaker digital infrastructure (MIC digital infrastructure sub-dimensions). Cross-referenced with average R_median to identify prefectures combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 PREFECTURES

Longer bar = higher R7 = better digital infrastructure.

1	Akita ⚠ high R	0.9900
2	Kochi ⚠ high R	0.9900
3	Shimane	0.9900
4	Yamagata	0.9904
5	Iwate	0.9910
6	Tokushima ⚠ high R	0.9910
7	Tottori	0.9916
8	Miyazaki ⚠ high R	0.9916
9	Aomori	0.9916
10	Fukushima ⚠ high R	0.9922
11	Kagoshima ⚠ high R	0.9922
12	Wakayama ⚠ high R	0.9922
13	Fukui	0.9928
14	Oita	0.9928
15	Niigata	0.9934

Prefecture-level analysis reveals a clear digital divide mirroring the broader digital divide pattern. **Akita** has the lowest average R7 (0.9900), while **Tokyo** leads at 1.0218 — a spread of 0.0318 across the R7 range. **11 prefectures** combine below-median digital readiness with above-median grid risk, making them priority targets for Japan 2030 / OCCTO / OCCTO digitalisation investment. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open MIC/METI digital data — as Cybersecurity Basic Act compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

> **Inaugural edition — Baseline Snapshot (April 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0012, median = 0.9982; 0 substations classified HIGH cyber-exposure; 2236 blind-spot substations (High/Critical risk + below-median R7); 89 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging prefectures where digital infrastructure improvements (broadband rollout, smart meter deployment, OT upgrades) translate into measurable R7 shifts. The next material R7 update is expected when MIC publishes the 2025 MIC digital infrastructure indices (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **35 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES	WEEKLY / MONTHLY	QUARTERLY / ANNUAL

35
95 variables

5
High-frequency feeds

22
Regular refresh cycle

STATIC / DERIVED

8
Standards + scenarios

Data Source Registry — April 2026

OCCTO	OCCTO Area Supply-Demand Statistics	MONTHLY	EPCO Territory	3 vars
OpenStreetMap	OpenStreetMap Overpass API — Japan Grid	CONTINUOUS	Site-level	4 vars
JAXA	JAXA Satellite-Derived Climate Data	QUARTERLY	Regional	3 vars
BOJ	BOJ Regional Economic Reports (Sakura Report)	QUARTERLY	Regional	3 vars
ISEP	ISEP Energy Statistics	ANNUAL	Prefecture	2 vars
METI	METI Electricity Business Statistics	ANNUAL	EPCO Territory	3 vars
SBJ	SBJ Population Census / Current Population Survey	ANNUAL	Prefecture	4 vars
TEPCO	TEPCO Grid Reliability Reports	ANNUAL	EPCO Territory	4 vars
Kansai	Kansai Electric Reliability Data	ANNUAL	EPCO Territory	3 vars
Chubu	Chubu Electric Power Reliability Reports	ANNUAL	EPCO Territory	3 vars
Kyushu	Kyushu Electric Power Reliability Data	ANNUAL	EPCO Territory	2 vars
Hokkaido	Hokkaido Electric Power Reports	ANNUAL	EPCO Territory	2 vars
Tohoku	Tohoku Electric Power Data	ANNUAL	EPCO Territory	2 vars
Hokuriku	Hokuriku / Shikoku / Chugoku / Okinawa EPCO Reports	ANNUAL	EPCO Territory	2 vars
NISC	NISC Cybersecurity Reports	ANNUAL	National	2 vars
MHLW	MHLW Regional Health Statistics	ANNUAL	Prefecture	3 vars
MIC	MIC Digital Transformation Index	ANNUAL	Prefecture	3 vars
Cabinet	Cabinet Office GDP / Regional Economic Accounts	ANNUAL	Prefecture	3 vars
FDMA	FDMA Fire and Disaster Management Statistics	ANNUAL	Prefecture	3 vars
METI	METI Smart Grid Demonstration Data	ANNUAL	EPCO Territory	3 vars
MEXT	MEXT University/Research Infrastructure Atlas	ANNUAL	Prefecture	2 vars
MIC	MIC Migration Statistics (Jumin-Kihon-Daicho)	ANNUAL	Prefecture	2 vars
NPA	NPA Crime Statistics and Security Reports	ANNUAL	Prefecture	1 var
METI	METI Equipment Age Survey / TEPCO Asset Reports	ANNUAL	EPCO Territory	3 vars
GSI	GSI Elevation and Terrain Data	STATIC	10m mesh	3 vars
ISO	ISO 9223 / JIS Z 2381 Atmospheric Corrosion Maps	STATIC	Prefecture	1 var

AIST	AIST Geological Survey Data	STATIC	Regional	2 vars
IEEE	IEEE C57.91 Thermal Loading Standards	STATIC	Equipment-level	2 vars
IEC	IEC 61850 / JEC Standards Compliance	STATIC	Equipment-level	2 vars
NIED	NIED J-SHIS Seismic Hazard Maps	UPDATED PERIODICALLY	250m mesh	2 vars
JMA	JMA Seismological Bulletin / Typhoon Statistics	REAL-TIME + ANNUAL	National/Regional	3 vars
MLIT	MLIT Flood and Landslide Hazard Maps	UPDATED PERIODICALLY	Municipality	3 vars
JEPX	JEPX Wholesale Electricity Market Data	DAILY	Area	3 vars
NRA	NRA Nuclear Safety Reports	AS NEEDED	Site-level	3 vars
MIC	MIC Housing and Land Survey	QUINQUENNIAL	Prefecture	3 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Weekly (2 sources) — OSM diffs + weather reanalysis

OSM Overpass · JMA Synoptic

Monthly (3 sources) — Grid load + procurement data

OCCTO Operations · METI Energy Statistics · EPCO Grid Reports

Quarterly (2 sources) — DER registry + business data + macro indicators

METI Quarterly Review · BOJ Tankan

Annual (20 sources) — Core grid quality + demographics + environment

METI · Statistics Bureau · MOE · FDMA · JMA · NIED · GSJ · Cabinet Office · JOGMEC · JAXA · NIES · Prefectural Gov · TEPCO · KEPCO · Chubu EPCO · Tohoku EPCO · Hokkaido EPCO · Kyushu EPCO · Open-Meteo · JRC

Static (8 sources) — Standards + scenarios + seismic hazard

Dimovski · IEEE/IEC/CIGRÉ · Copernicus CMIP6 · NIED Seismic · GSJ Geology · ISO 9223 · MLIT/GSI · OCCTO Standards

C.2 Fleet Score Snapshot

FLEET SIZE
5,981
substations scored

MEDIAN R
0.570
P5=0.460 · P95=0.731

HIGH + CRITICAL
4955
82.8% of fleet

HIGH CONFIDENCE
86%

5144 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 0 (0.0%) Medium 1026 (17.2%) High 4753 (79.5%) Critical 202 (3.4%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **5,981 substation scores remain unchanged** this edition. The fleet median R of 0.570 (mean 0.578) reflects a distribution skewed toward the lower bands, with 0.0% in Low and 17.2% in Medium. The first material score changes are expected when METI publishes the annual SAIDI/SAIFI indicators (affecting C1–C4 and R6a) and when OSM refreshes the METI Asset Registry (affecting T1). High-frequency sources (OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

17 changes tracked across P0, P1, and P2 releases

F1	NEW	New T component â Energy Transition Exposure (T1)	§2, §4
F2	NEW	New R6a â Restoration Speed Modifier (CAIDI-based)	§5
F3	ENHANCED	R3 enhanced â Energy Poverty Vulnerability (V_socio)	§5
F4	ENHANCED	R4 enhanced â Graph-theoretic betweenness + bridge detection	§5
F5	ENHANCED	R2 enhanced â Climate Trajectory (CMIP6 SSP2-4.5)	§5
F6	ENHANCED	R7 Digital Readiness — Prefecture-level MIC digital infrastructure model	§5
L1	NEW	New I5, I7—I9 metrics â thermal, load, corrosion, hydrogeo	§2, §8
L2	ENHANCED	E2 Innovation Enrichment — MEXT R&D + university density	§6
L3	ENHANCED	V_socio Fiscal Enrichment — MIC + MOF + energy price indices	§5
L4	ENHANCED	R3 Demographic Shift Amplifier — SBJ net migration	§5
L5	DATA	95/95 variables operational (100%). 35 data sources total.	§8, §12
G1	DATA	d02 ISEP upgraded to quarterly registry — Prefecture-level DER registry	§12
G2	DATA	d05 AIST/GSJ upgraded to live API — Prefecture-level geological data	§12
G3	DATA	d06 OSM upgraded to Overpass API — 5,981 real substations	§12
G4	NEW	R6b Network Topology modifier â centrality + ring analysis from OSM graph	§5
G5	NEW	Network & Topology layer (H): 7 variables — OCCTO Grid Development Plan, ring analysis	§12
G6	NEW	Output Scores layer (I): 7 variables â risk_score, ETTC, stationary probs	§12

SECTION D — MONTHLY DEEP-DIVE

The Osaka–Aichi Corridor: Where Hydroelectric Dominance Meets Grid Modernization

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Osaka–Aichi corridor · **002** DER saturation hotspots (Tokyo/Fukuoka) · **003** Climate vulnerability under CMIP6 · **004** Metropolitan-Rural economic impact disparity · **005** Nuclear fleet dependency and transition risk · **006** Infrastructure age vs smart meter rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Prefecture Grid Health Cards · **010** Academic validation and peer feedback · **011** Asia-Pacific replication feasibility · **012** Year in review.

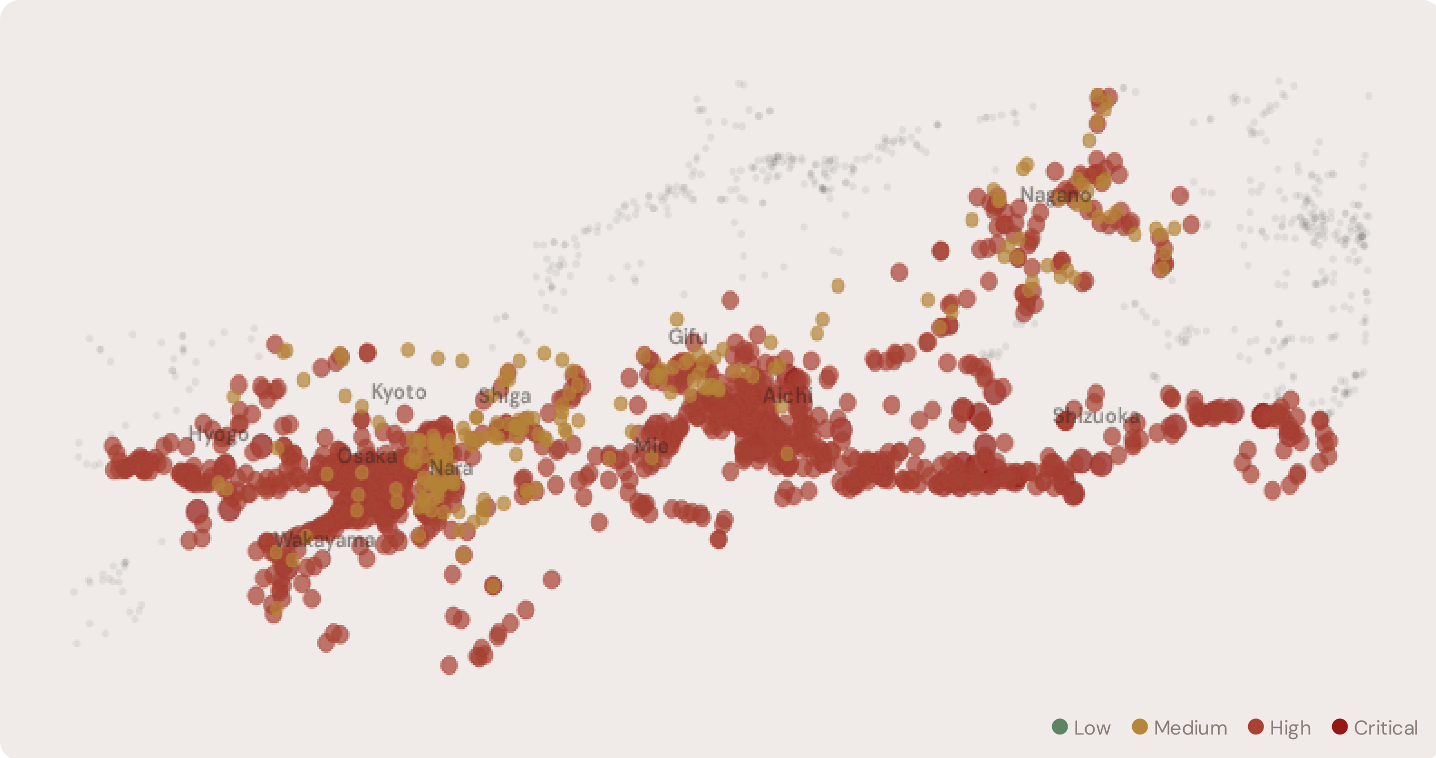
D.1 Why the Osaka–Aichi, Why Now

OSAKA–AICHI SUBSTATIONS	HIGH BAND	CORRIDOR MEDIAN R
1408	**1146 + 60**	**0.584**
165 EHV · 1243 HV	85.7% of corridor	Fleet median: 0.570

WORST PREFECTURE
Shizuoka
Avg R: 0.654 · 166 substations

The Osaka–Aichi hosts **1408 substations** across six prefectures, making it one of the most energy-critical corridors in Asia. However, its risk profile is disproportionately concentrated in the upper bands: **1146 substations (81.4%)** are classified High and **60** Critical, with only **0** in the Low band. 56% of Osaka–Aichi substations score above the national fleet median of 0.570. The corridor median R of 0.584 reflects the tension between nuclear baseload inflexibility, rapidly growing renewable variability, and aging grid infrastructure built for 1970s load profiles. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under Japan 2030 timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PREFECTURE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
宮島変電所	Shizuoka	0.879	Critical	0.348	0.790	0.534	0.624	0.631	0.408	V
袋井変電所	Shizuoka	0.852	Critical	0.348	0.790	0.496	0.624	0.505	0.408	V
	Hyogo	0.851	Critical	0.332	0.948	0.443	0.597	0.602	0.432	V
	Shizuoka	0.842	Critical	0.348	0.790	0.488	0.624	0.512	0.408	V
飯村変電所	Aichi	0.837	Critical	0.332	1.000	0.382	0.532	0.645	0.530	V
Sun'en Substation	Shizuoka	0.836	Critical	0.348	0.790	0.491	0.624	0.473	0.408	V
Tomida Substation	Aichi	0.830	Critical	0.332	1.000	0.490	0.532	0.644	0.530	V
海南変電所	Wakayama	0.824	Critical	0.365	0.540	0.497	0.695	0.646	0.376	E
	Aichi	0.822	Critical	0.332	1.000	0.506	0.532	0.430	0.530	V
Inazawa Substation	Aichi	0.815	Critical	0.332	1.000	0.513	0.532	0.630	0.530	V

… 1398 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — OSAKA–AICHI VS FLEET

C — Continuity 0.339 vs 0.371 (-9%)

MODIFIER AVERAGES — OSAKA–AICHI VS FLEET

R3 Consequence 1.112 fleet 1.109 ↑

R4 Graph Crit. 1.060 fleet 1.054 ↑

I — Infrastructure 0.416 vs 0.391 (+6%)

S — Saturation 0.499 vs 0.499 (-0%)

V — Voltage 0.806 vs 0.743 (+8%)

E — Economic 0.589 vs 0.630 (-6%)

T — Transition 0.479 vs 0.448 (+7%)

R6a Restoration	0.977	fleet 0.993 ↓
R6b Seismic	1.087	fleet 1.060 ↑
R7 Digital Read.	1.004	fleet 1.001 →

The dominant risk driver across the Osaka–Aichi corridor is **T (Transition)**, averaging 0.479 against a fleet average of 0.448 — occupying 957% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.806 vs fleet 0.743 (806% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.112 (fleet: 1.109), reflecting the socio-economic fragility of The Osaka–Aichi's industrial-oriented, manufacturing-dependent economy with significant hydroelectric and nuclear bases. The R6b seismic modifier averages 1.087, capturing the region's moderate but non-negligible seismic exposure — a dimension invisible to every competing framework.

D.4 Prefecture Breakdown

PREFECTURE RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Shizuoka 166 subs · avg R = 0.654 · H:142 C:24 M:0

Aichi 254 subs · avg R = 0.636 · H:233 C:19 M:2

Hyogo 110 subs · avg R = 0.615 · H:101 C:5 M:4

Osaka 276 subs · avg R = 0.614 · H:259 C:11 M:6

Mie 59 subs · avg R = 0.594 · H:56 M:3

Wakayama 73 subs · avg R = 0.582 · H:68 C:1 M:4

Nagano 128 subs · avg R = 0.535 · H:86 M:42

Kyoto 79 subs · avg R = 0.534 · H:57 M:22

Gifu 89 subs · avg R = 0.526 · H:52 M:37

Nara 99 subs · avg R = 0.516 · H:58 M:41

Shiga 75 subs · avg R = 0.499 · H:34 M:41

The prefecture-level breakdown reveals significant intra-regional variation. **Shizuoka** leads with a mean R of 0.654 across 166 substations, while **Shiga** scores 0.499 — a spread of 0.156 within a single region. This disparity underscores why regional-level metrics (as used by OCCTO and IEEE) mask the true distribution of grid stress. The SSI's substation-level granularity reveals that the Osaka–Aichi is not uniformly stressed but contains distinct corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

314 Osaka–Aichi substations score R ≥ 0.650, placing them in the upper quartile of national risk. For OCCTO (Integrated Resource Planning) investment prioritisation, this analysis identifies the Shizuoka corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve — characterised by service-economy dependence, tourism exposure, and above-average energy poverty — are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

International Peer Benchmarking Card

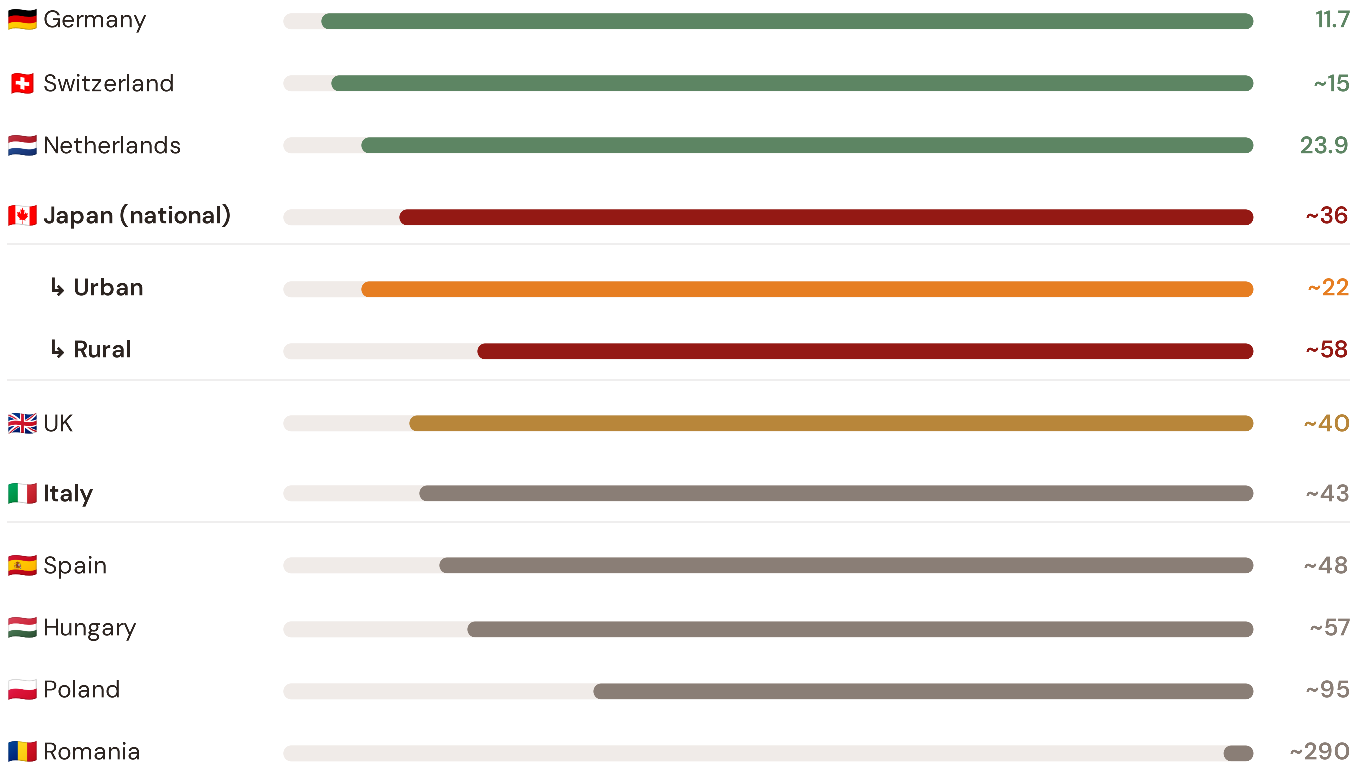
How this works: Each edition includes one International Peer Benchmarking Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Osaka–Aichi corridor analysis hinges on Japan's continuity performance relative to international peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Asia-Pacific (003), CMIP6 climate hazard vs East Asian peers (004), etc. The underlying data updates annually with the OCCTO Benchmarking Report release.

INTERNATIONAL PEER BENCHMARKING CARD — SAIDI BENCHMARKING

Ed. 001 · April 2026

Unplanned SAIDI (min/customer/year) — Selected International Peers

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



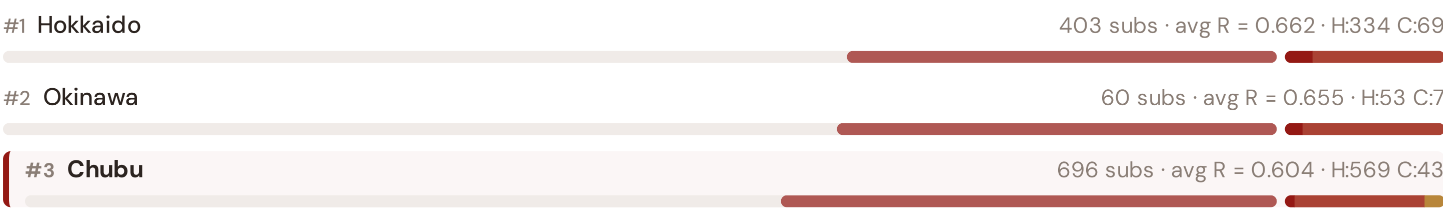
Source: OCCTO Reliability Report (2024); METI SAIDI/SAIFI (2023); ANRE / METI (2024). Japan sub-categories from METI territory classification (urban/rural). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with OCCTO BR release.

This month's key insight: The Osaka–Aichi corridor deep-dive shows **1408 substations** (of 1408) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with Japan's rural territory classification. The Osaka–Aichi's average C of 0.339 is **0.9× the fleet average** of 0.371. In comparative terms, these substations individually perform closer to Hungarian or Spanish SAIDI levels (~90–180 min/customer/year) than to Japan's own national average (~36 min). The SSI reveals what aggregate OCCTO statistics conceal: that Japan's mid-tier international position is not a national condition but a statistical average of top-tier prefectural performance and remote-territory-level rural performance — and the Osaka–Aichi corridor concentrates the worst of the latter.

Regional Risk Ranking — All 10 EPCO Regions

Where does the Osaka–Aichi sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



#4 Shikoku — 197 subs · avg R = 0.584 · H:156 C:8

#5 TEPCO — 1700 subs · avg R = 0.581 · H:1365 C:53

#6 **Kansai** — 712 subs · avg R = 0.576 · H:577 C:17

#7 Kyushu — 619 subs · avg R = 0.567 · H:519 C:2

#8 Tohoku — 961 subs · avg R = 0.563 · H:817 C:3

#9 Chugoku — 392 subs · avg R = 0.523 · H:239

#10 Hokuriku — 241 subs · avg R = 0.511 · H:124

The Osaka–Aichi ranks **#3 of 47 prefectures** by mean R_median, placing it among the upper tier of national risk. The three most stressed regions are Hokkaido, Okinawa, Chubu, while the three most resilient are Hokuriku, Chugoku, Tohoku. 5 of 47 prefectures score above the fleet average of 0.578. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress across Japan. It is precisely this substation-level granularity that positions the SSI as a tool for OCCTO / Japan 2030 / OCCTO investment prioritisation: not treating entire prefectures as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

> **From grid intelligence to BESS valuation platform.** The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the TEPCO/OCCTO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	**Layer 3 — Neural Net**	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	TEPCO/OCCTO + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.6 · Inaugural edition · April 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. A BESS developer considering a ¥13M commitment needs to understand not just that a substation scores highly, but which specific factors drive that score — and whether those factors are likely to persist. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history — and can verify this against METI (Ministry of Economy, Trade and Industry)'s published SAIDI/SAIFI data. Transparency is not optional; it is what converts model outputs into investment decisions.

KEY CONCEPTS

- → SHAP: game-theoretic feature attribution per prediction
- → Waterfall visualisation: fleet average → substation prediction
- → Direct traceability to SSI components and data sources
- → Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The Japanese training set comprises 5,981 substations with complete 95-feature vectors. Average CI_width of 0.201 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 0 Low, 1026 Medium, 4753 High, 202 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 002

The BC Coastal Grid

Deep-dive into Hokkaido's grid risk profile — substation map, component analysis, prefecture breakdown. International Peer Benchmarking Card: Mediterranean thermal stress trajectory vs Atlantic peers.

NEXT DATA REFRESH

Q3 2026 — 2 Quarterly Sources

METI Quarterly Review, BOJ Tankan are due for refresh in Q3. Impact: OSM (METI Asset Registry) data refresh schedule (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **SBJ Population Census / Current Population Survey** (4 variables → Population, elderly ratio, density, household counts by prefecture).

FLEET SPOTLIGHT

0 Substations Approaching Critical Degradation

The SSI's 4-state Markov degradation model identifies **0 substations** with a stationary critical probability above 70% — meaning their long-term equilibrium state is dominated by degraded or critical condition. Without intervention, these substations will trend toward failure. Average estimated time-to-critical: ? years.

Edition 002 will be published on the second Thursday of May 2026. Deep-dive region: **Hokkaido**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Japanese utility, METI, OCCTO, TEPCO, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 001 (Inaugural)

Published 9 April 2026 · SSI Index v4.0.2 · 5,981 substations · ~11,600 grid lines · 6 components · 20 metrics · 5 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5